

p2_g2_report

2026-02-19

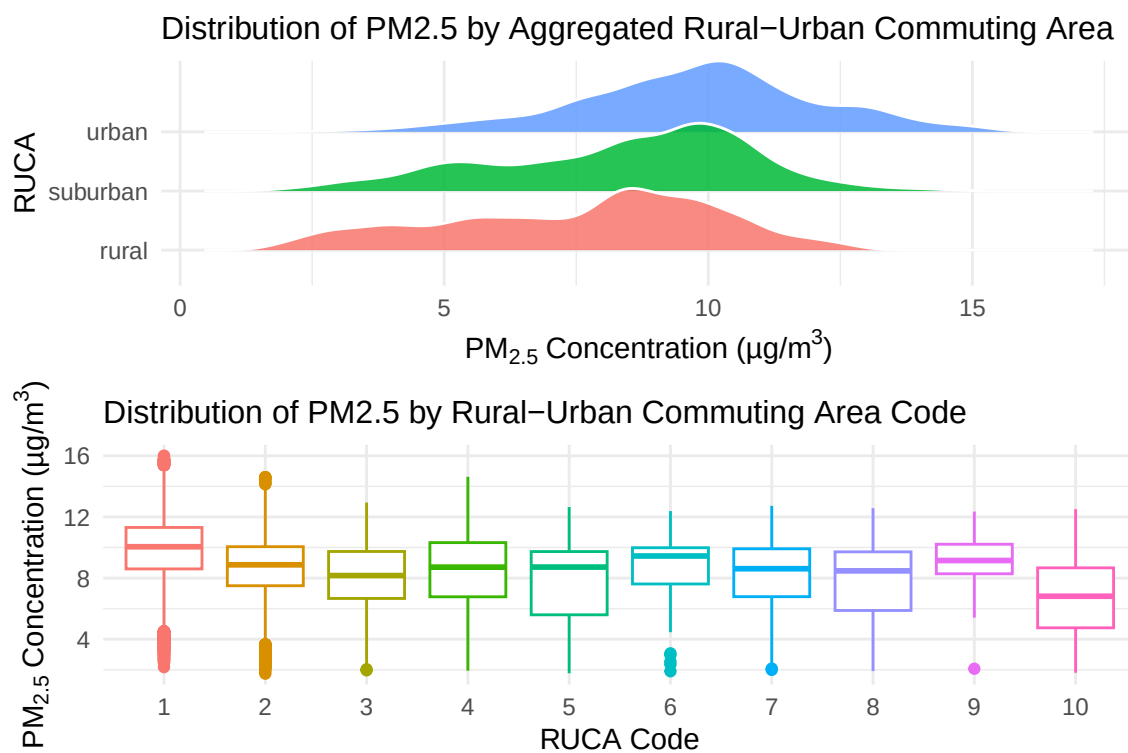
Appendix

EDA Section

Table 1: Census Tract Characteristics, by RUCA Categorization – Median (Q1, Q3)

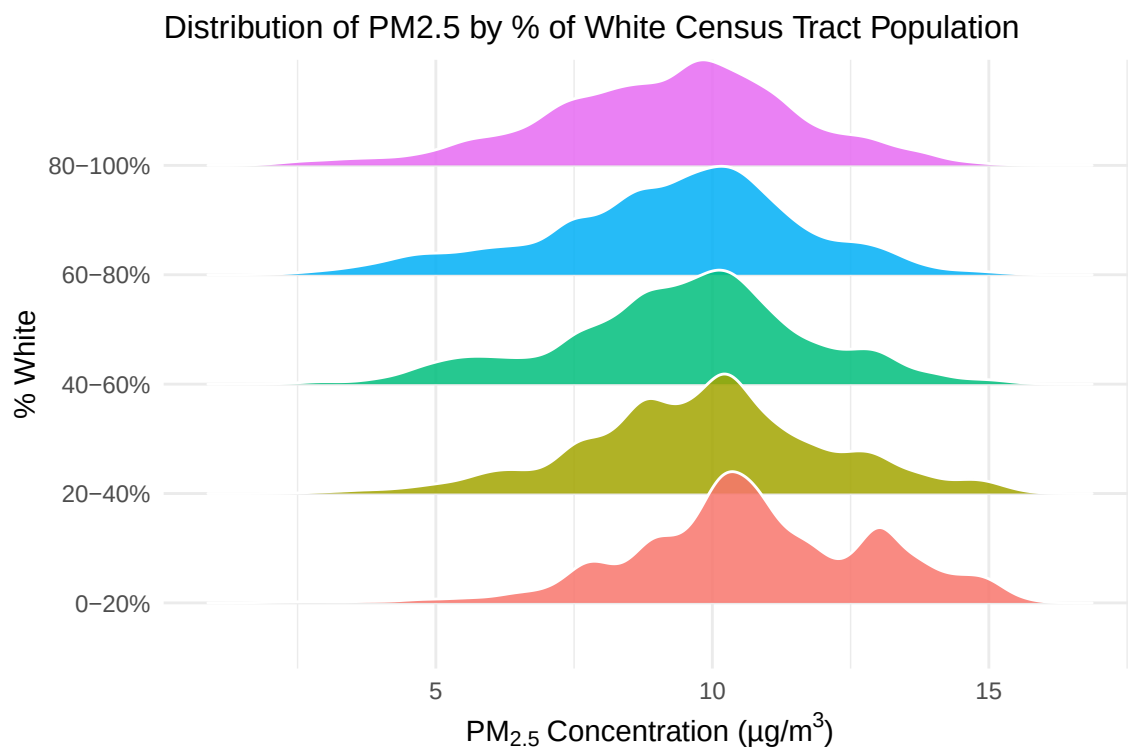
Characteristic	Overall N = 58,030	rural N = 1,126	suburban N = 1,702	urban N = 55,202	p- value
PM 2.5	9.87 (8.34, 11.10)	8.12 (5.60, 9.55)	8.76 (6.66, 10.12)	9.92 (8.45, 11.17)	<0.001
Coal Power Plant Impact (HyADS)	0.67 (0.22, 1.03)	0.65 (0.25, 1.00)	0.72 (0.37, 1.10)	0.67 (0.22, 1.02)	<0.001
Black Carbon	9.00 (8.11, 10.50)	8.17 (7.39, 9.18)	8.43 (7.80, 9.53)	9.04 (8.15, 10.60)	<0.001
Total Population	4,098.00 (2,971.00, 5,431.00)	3,753.50 (2,726.00, 4,950.00)	4,105.50 (3,065.00, 5,420.00)	4,104.50 (2,973.00, 5,441.00)	<0.001
Population Density (per km²)	1,110.85 (328.46, 2,302.84)	25.80 (9.78, 79.39)	137.35 (37.07, 545.48)	1,185.90 (403.27, 2,390.12)	<0.001
Percent in Poverty	10.65 (5.29, 19.81)	13.50 (9.04, 19.37)	13.71 (8.78, 20.60)	10.44 (5.16, 19.79)	<0.001
Percent White	69.67 (36.67, 86.57)	89.27 (70.46, 95.78)	84.49 (66.61, 93.61)	68.60 (35.04, 85.87)	<0.001
Percent Black	4.61 (1.38, 15.92)	0.63 (0.28, 3.59)	1.50 (0.49, 7.00)	4.83 (1.50, 16.46)	<0.001
Percent Asian	1.92 (0.67, 5.06)	0.41 (0.22, 0.74)	0.62 (0.30, 1.24)	2.07 (0.73, 5.34)	<0.001
Percent Hispanic	7.54 (3.02, 21.32)	2.95 (1.43, 9.31)	4.69 (1.73, 13.45)	7.79 (3.15, 21.86)	<0.001

Figure 1: Distribution of PM2.5 by Rural-Urban Commuting Area and Code



For the below, can include either the combined figure or just the PM_{2.5} figure!

Figure 2: Distribution of PM_{2.5} by % of White Census Tract Population



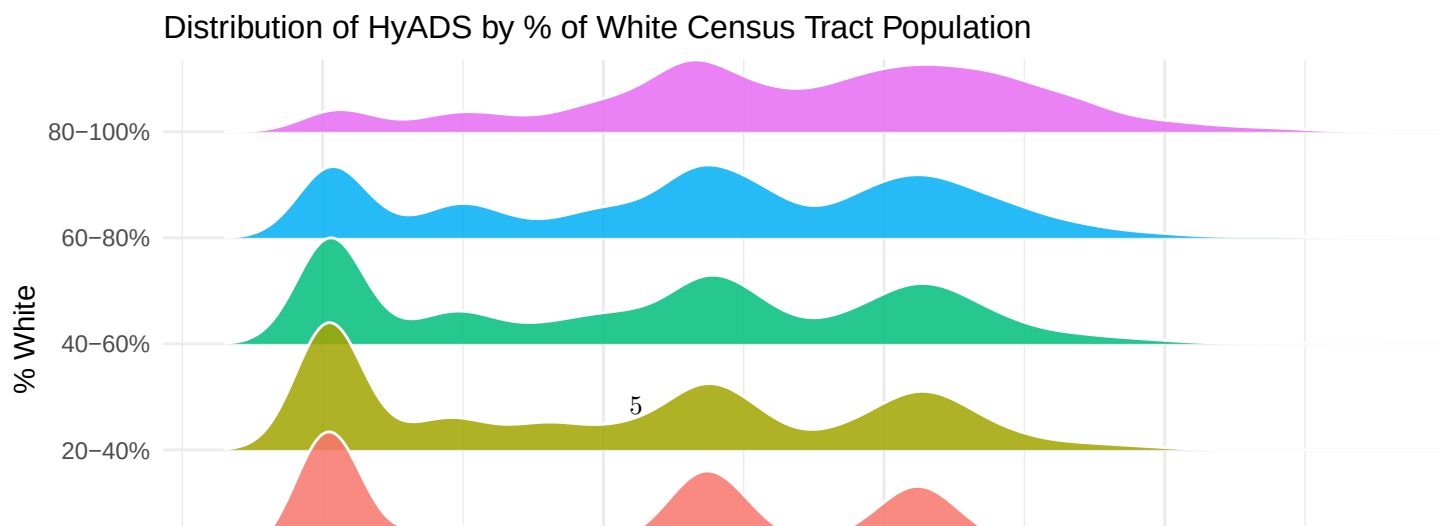
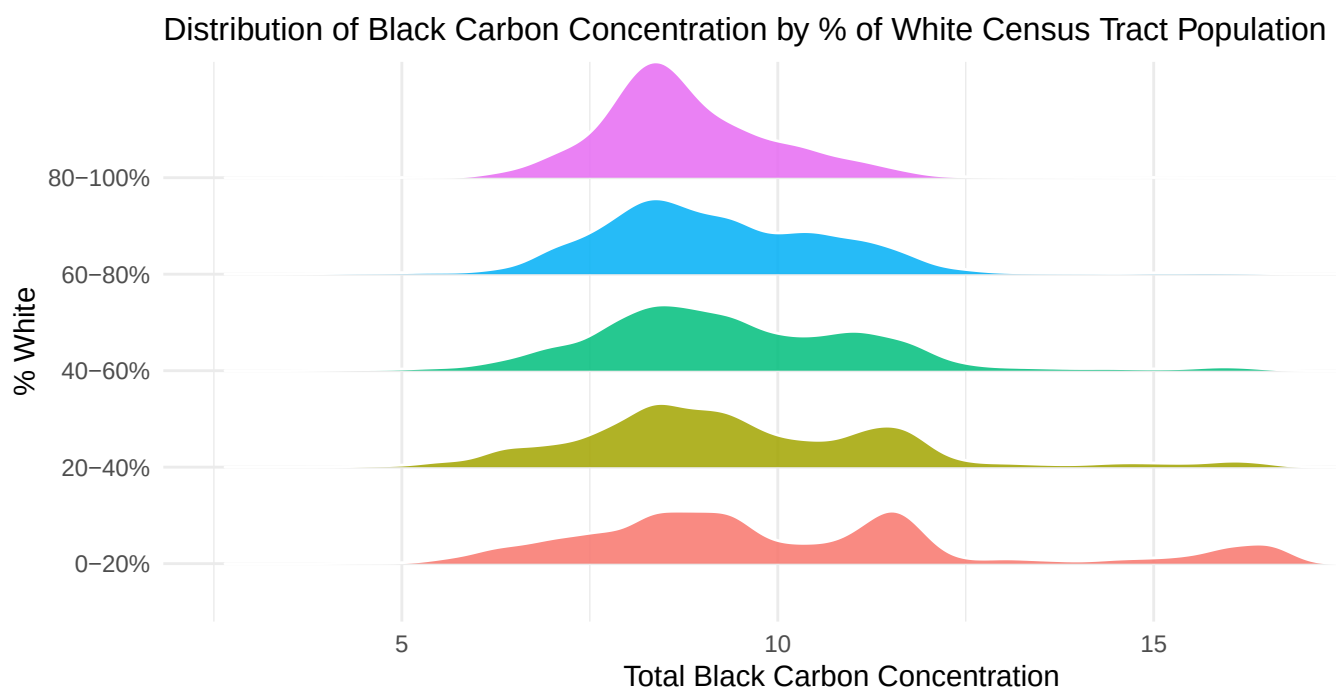
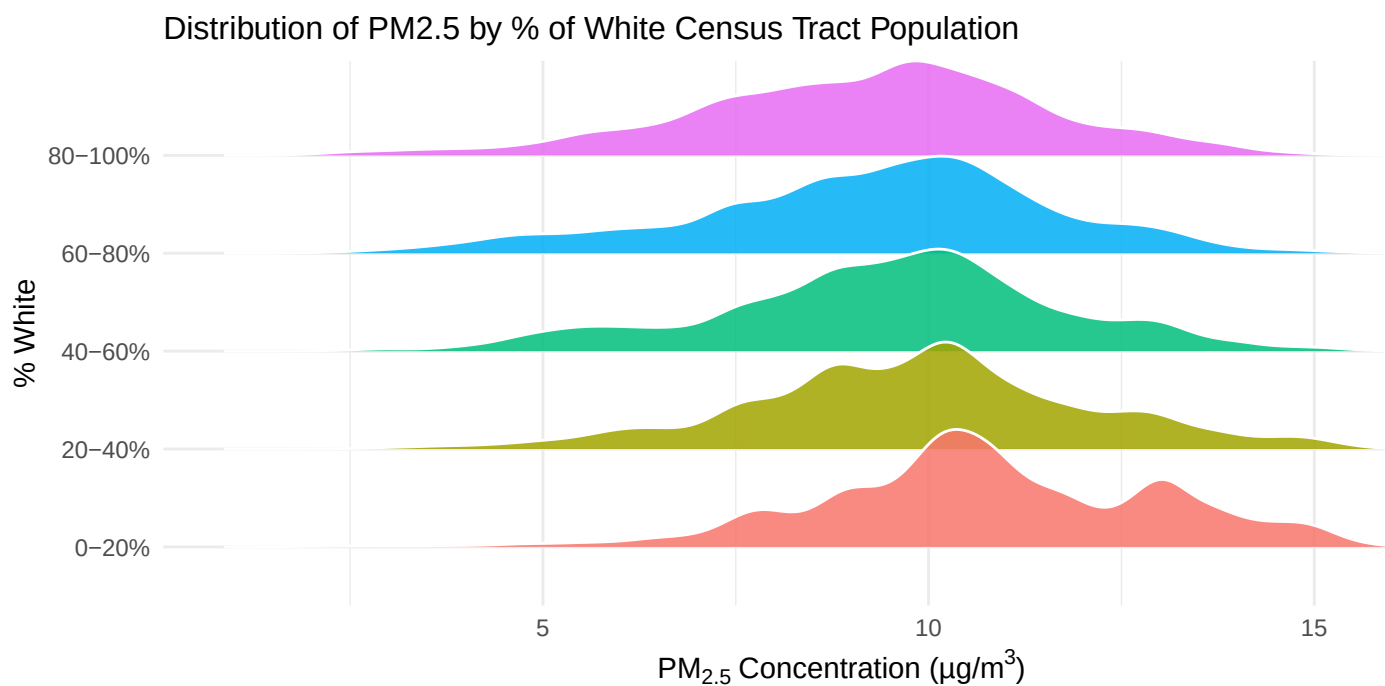


Figure 3: MAPS I will add them

Figure 4: Racial Composition by PM2.5 Exposure Percentile

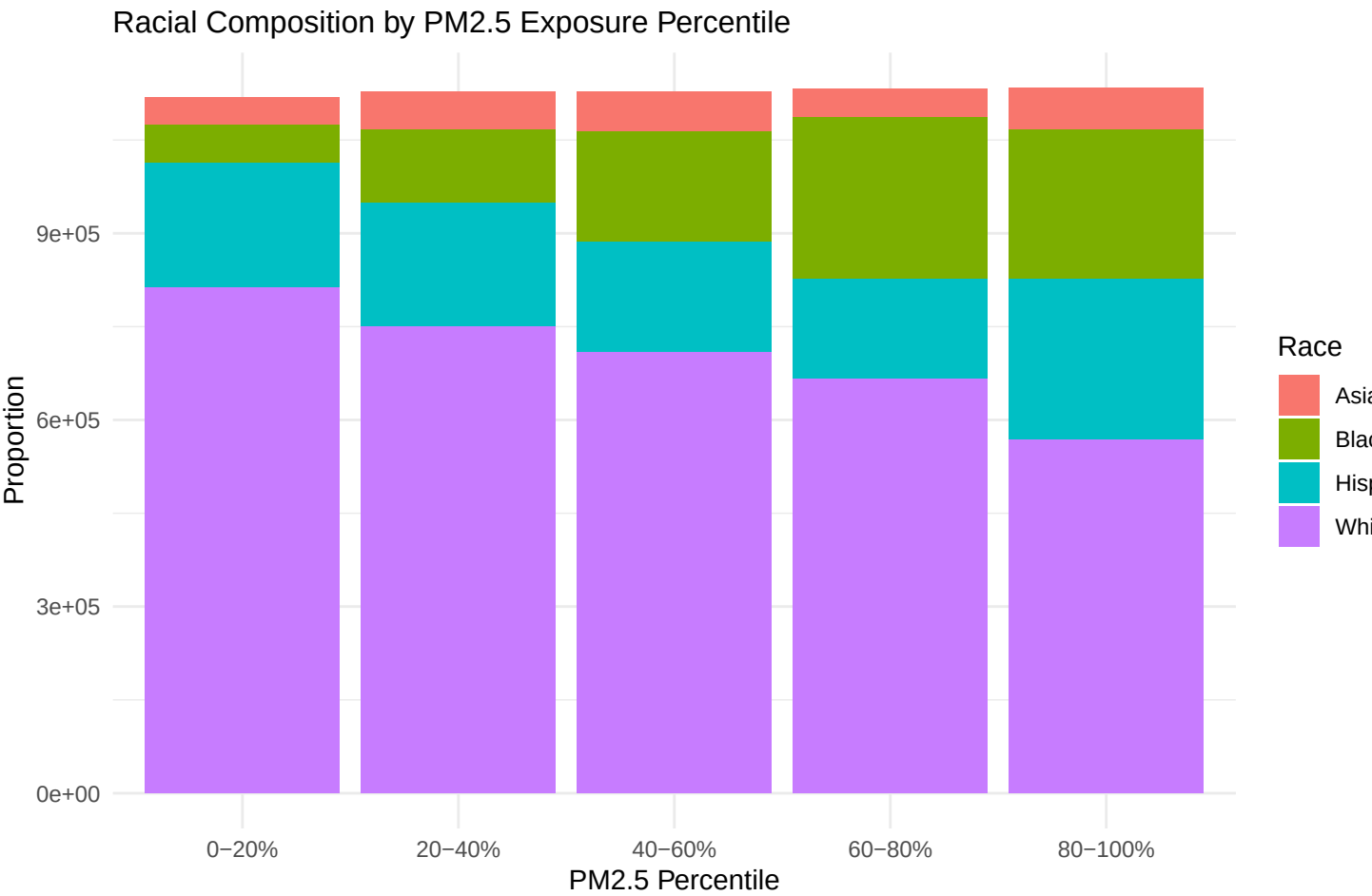
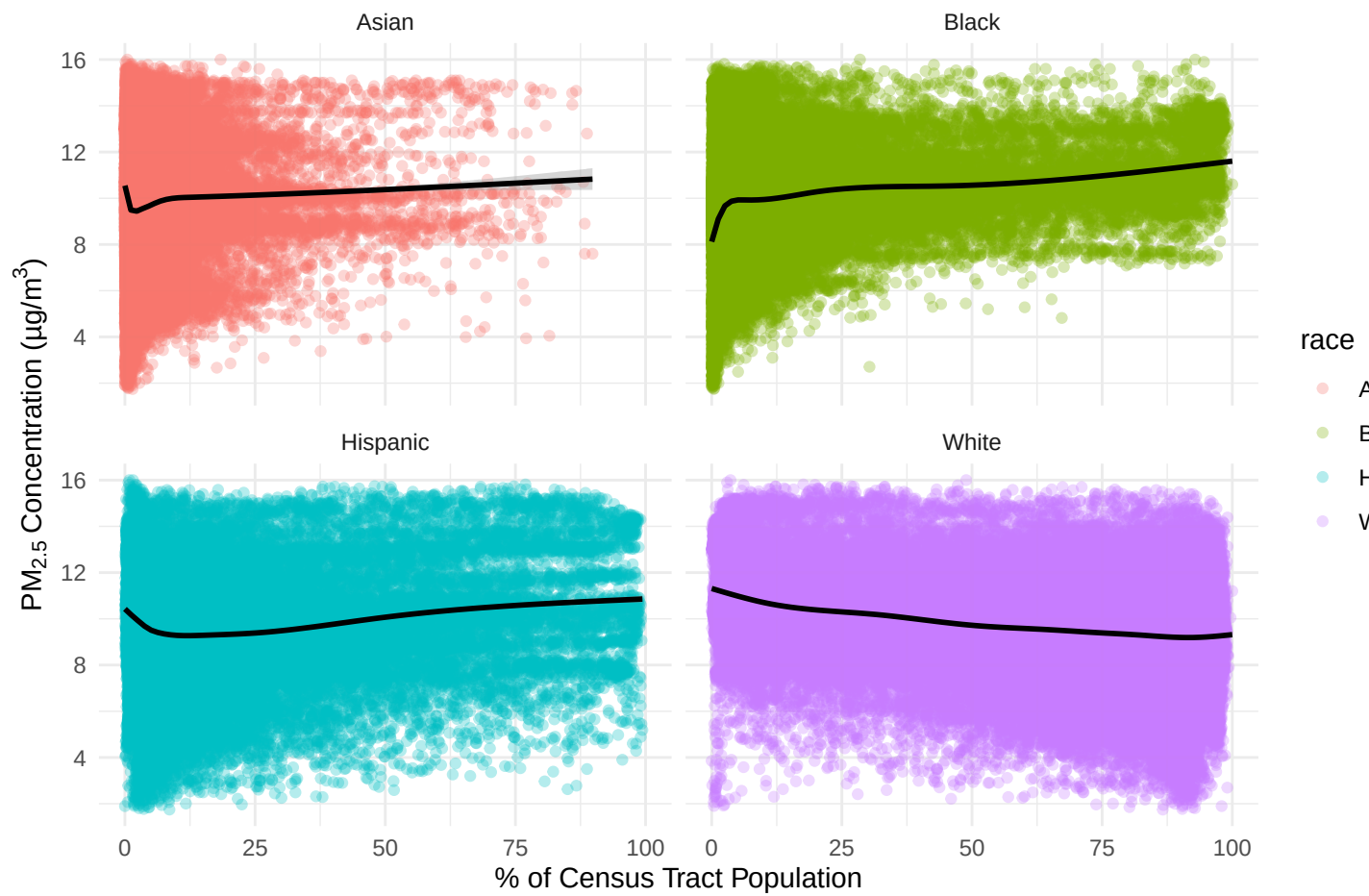
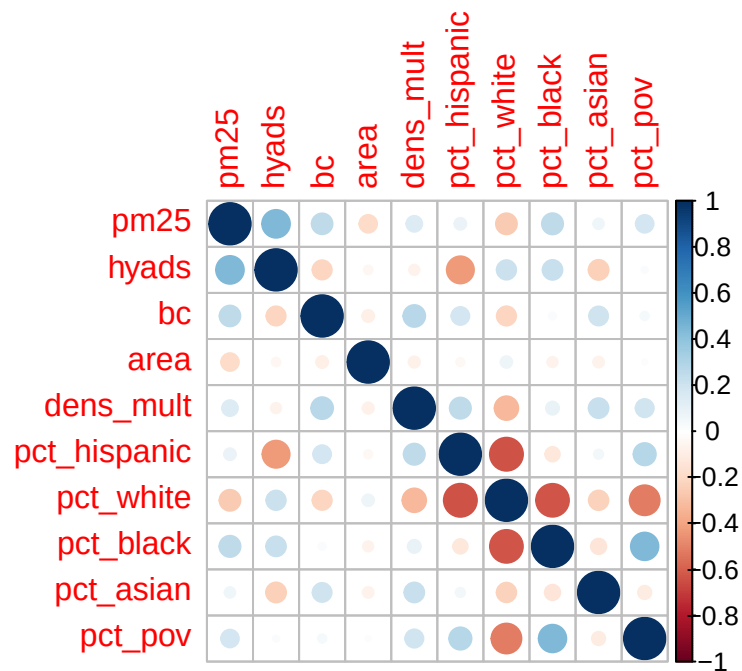


Figure 5: PM2.5 vs. of % Race of Census Tract Population



EDA: Variable Selection

Correlation



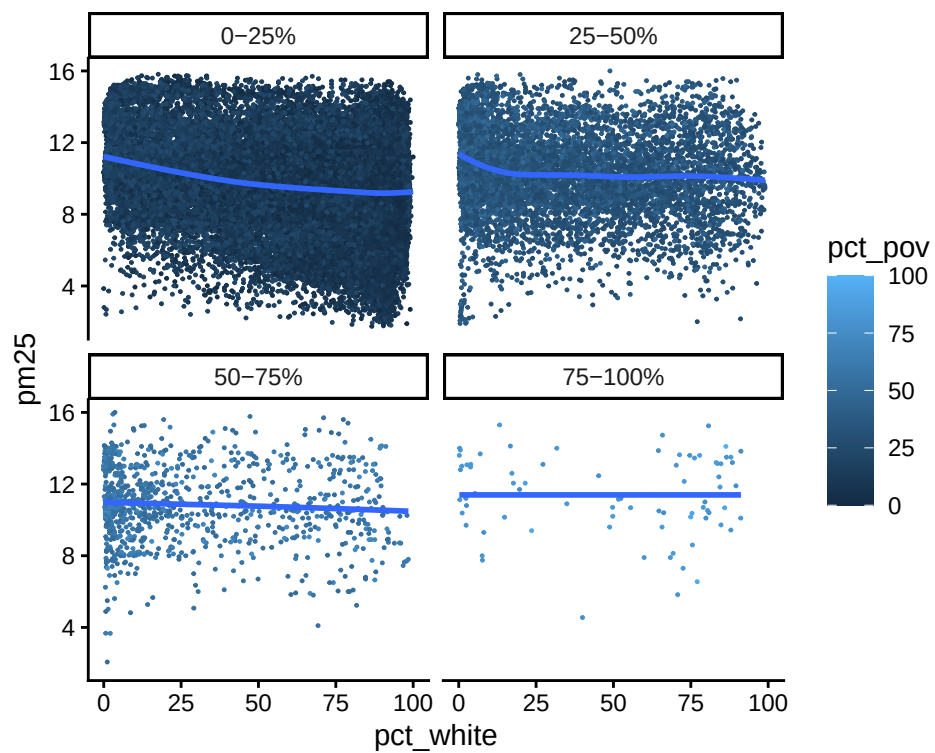
POV VS WHITE VS PM2.5

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'

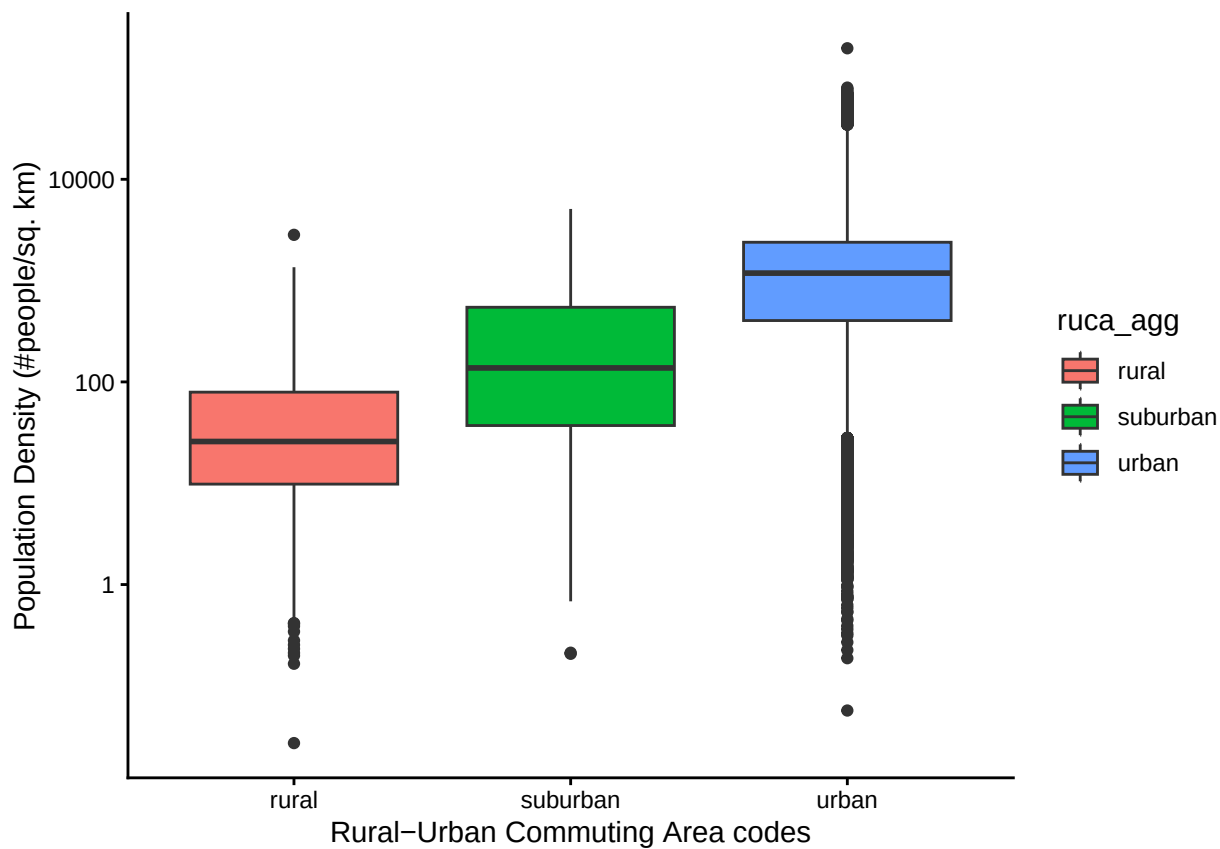
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
## The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
## The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
## The following aesthetics were dropped during statistical transformation:
## colour.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



RUCA, density and PM2.5



Modeling

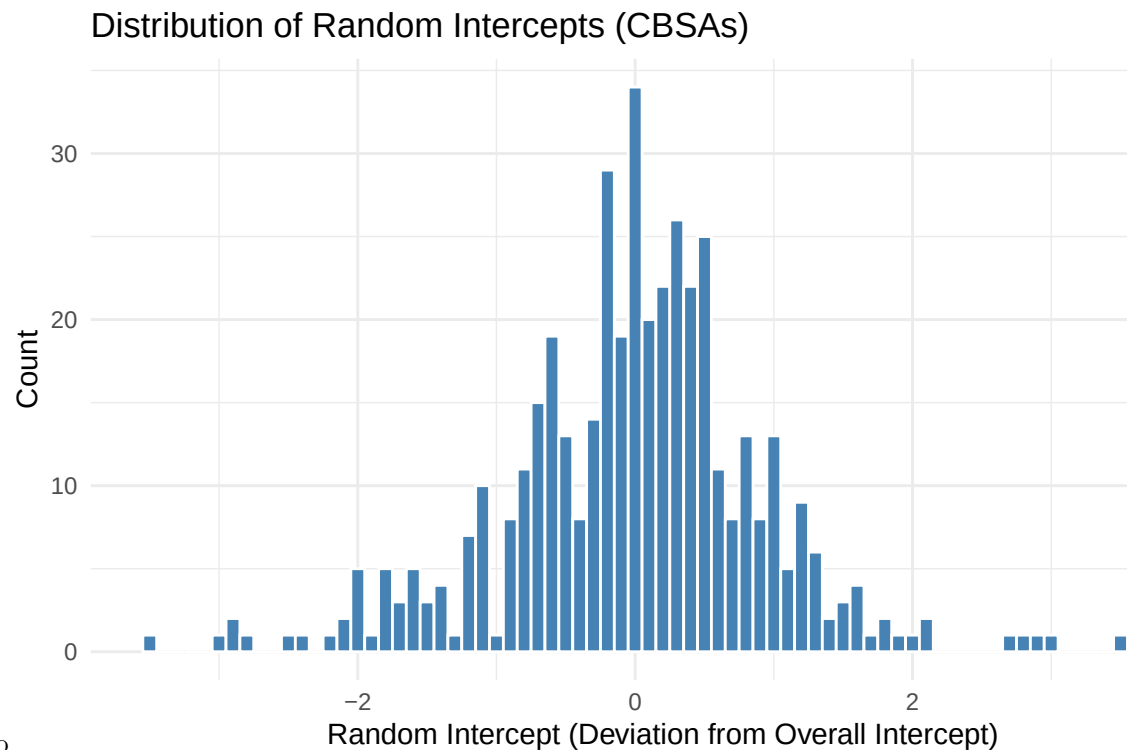
Approach 1: Nested LMM

nobs	sigma	logLik	AIC	BIC	REMLcrit	df.residual
58029	0.7863053	-69506.25	139032.5	139122.2	139012.5	58019

Characteristic	Beta	95% CI	p-value
(Intercept)	6.510	5.895, 7.125	<0.001
pct_white	-0.007	-0.007, -0.007	<0.001
pct_pov	-7.39e-04	-0.002, 3.13e-04	0.2
ruca_agg			
rural	—	—	
suburban	-0.236	-0.299, -0.172	<0.001
urban	0.370	0.318, 0.422	<0.001
log(dens_mult)	0.415	0.410, 0.420	<0.001
pct_white * pct_pov	1.10e-04	9.17e-05, 1.28e-04	<0.001

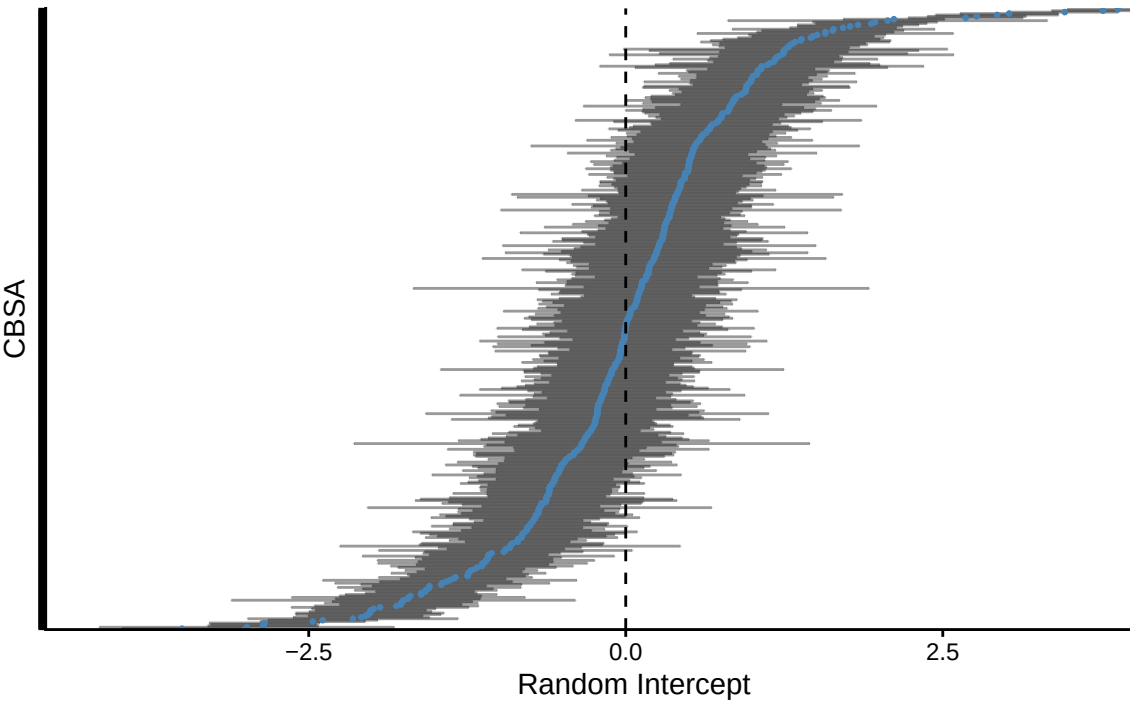
Abbreviation: CI = Confidence Interval

Note for Interpretation: per 10% increase



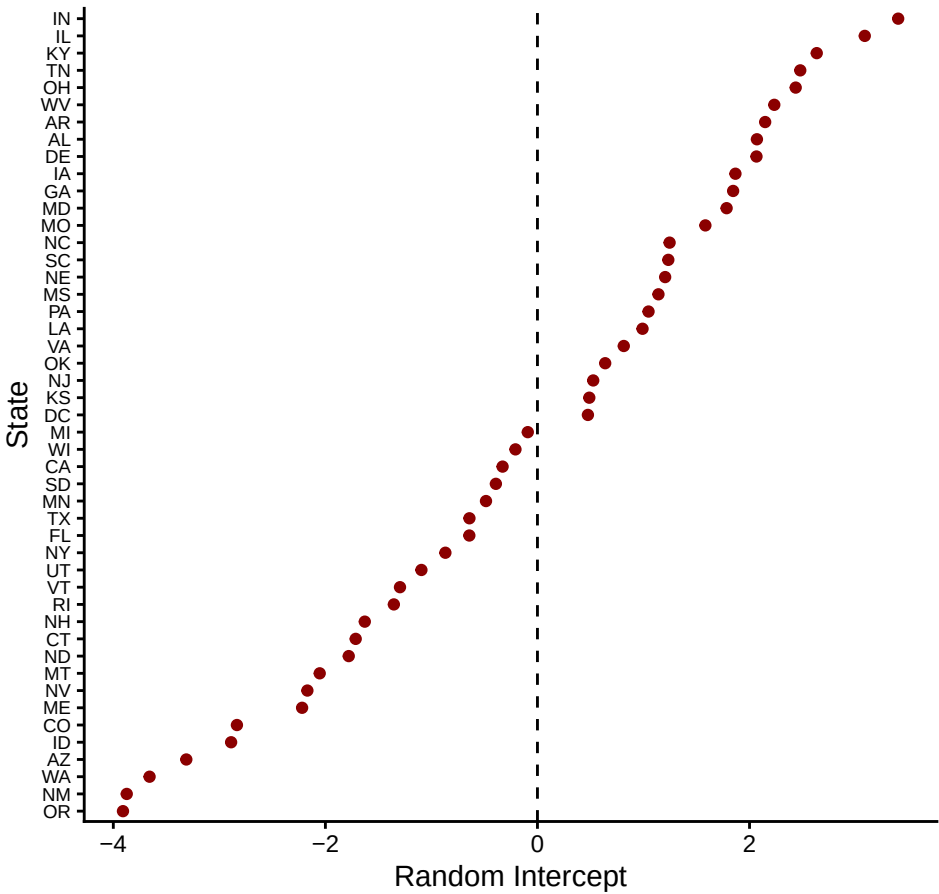
Checking random effects too

Random Intercepts with 95% CI (CBSA)



CBSA RI within each state

Random Intercepts by State

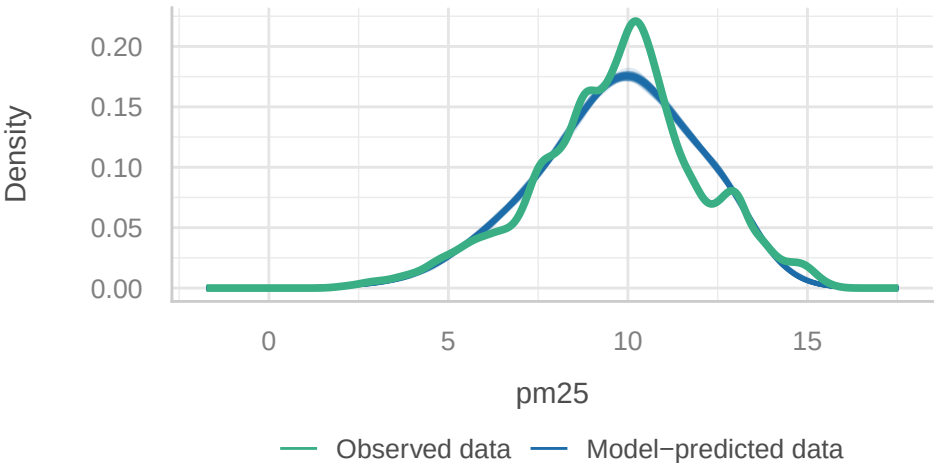


State-level RI

Model Diagnostics

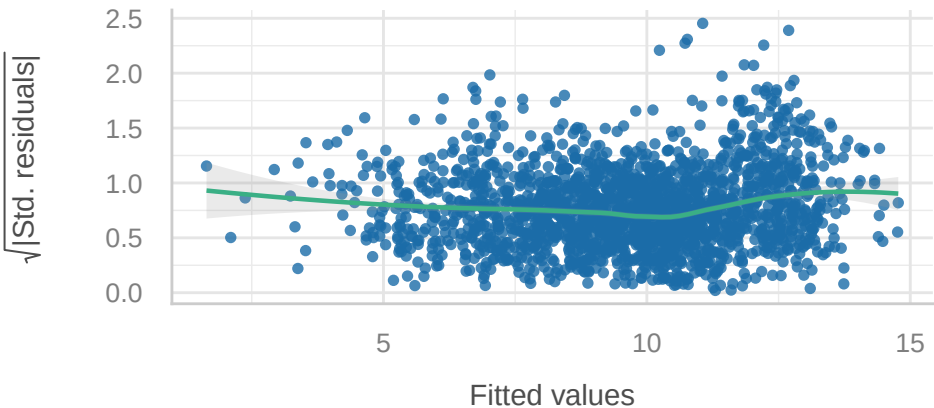
Posterior Predictive Check

Model-predicted lines should resemble observed data line



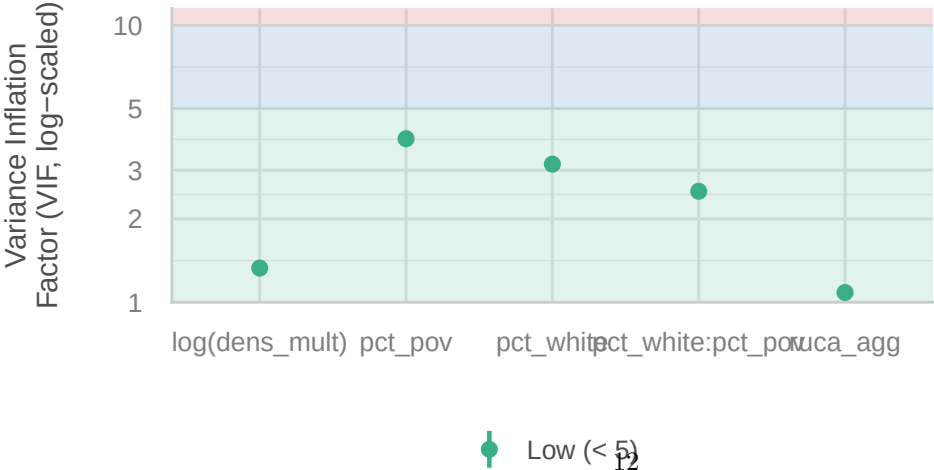
Homogeneity of Variance

Reference line should be flat and horizontal



Collinearity

High collinearity (VIF) may inflate parameter uncertainty

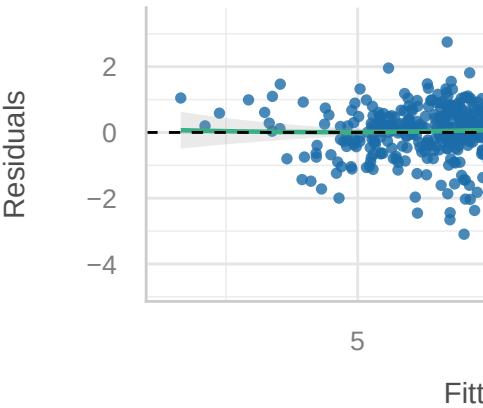


Normality of Random Effects (cbsa_id:state_abb)

Dots should be plotted along the line

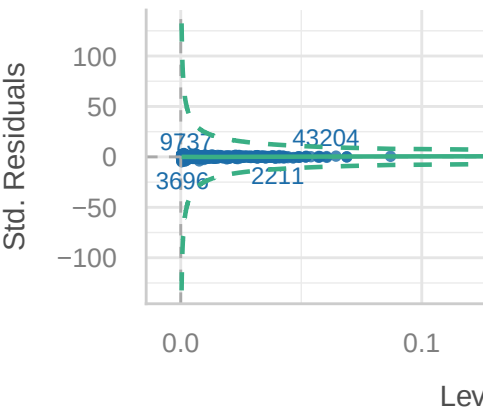
Linearity

Reference line should be flat and horizontal



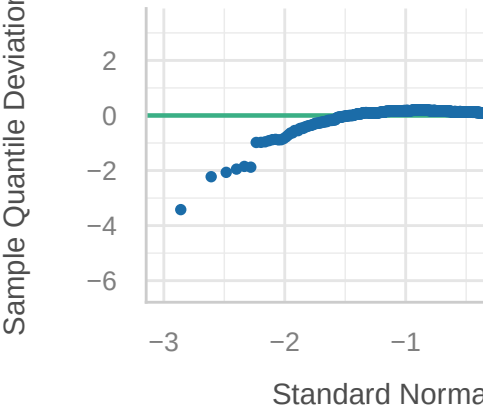
Influential Observations

Points should be inside the contour line



Normality of Residuals

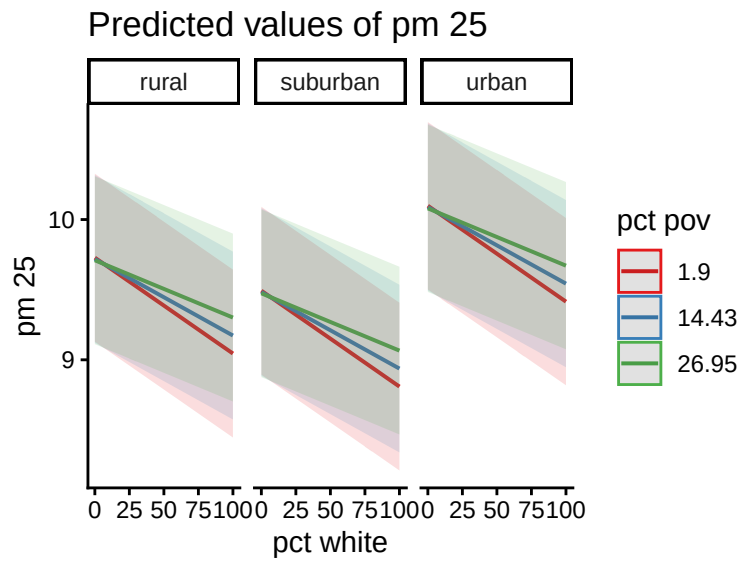
Dots should fall along the line



Normality of Random Effects (state_abb)

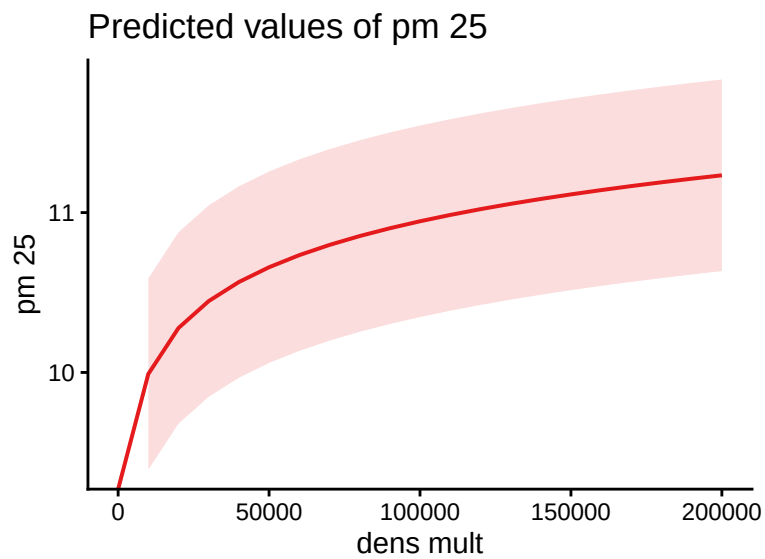
Dots should be plotted along the line

Prediction



Prediction for log(dens)

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_ribbon()`).
```



Approach 2: Cross classified Multilevel modeling (CCMM)

```
## [1] 47
```

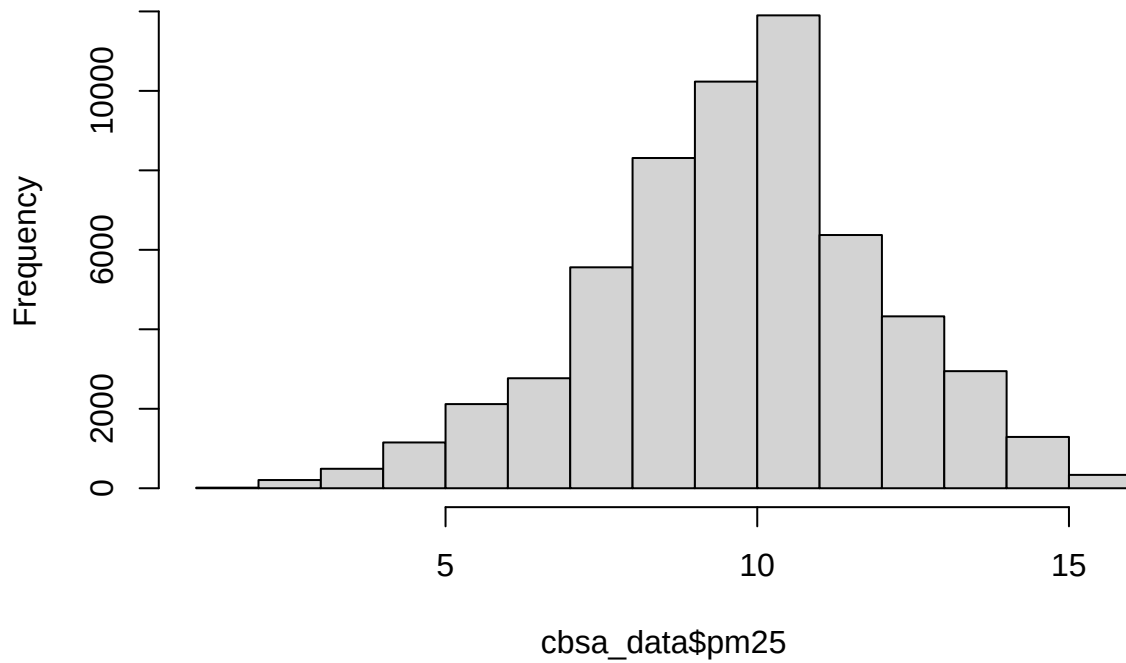
```
## [1] 381
```

```
## [1] 1119
```

```
## [1] 58030
```

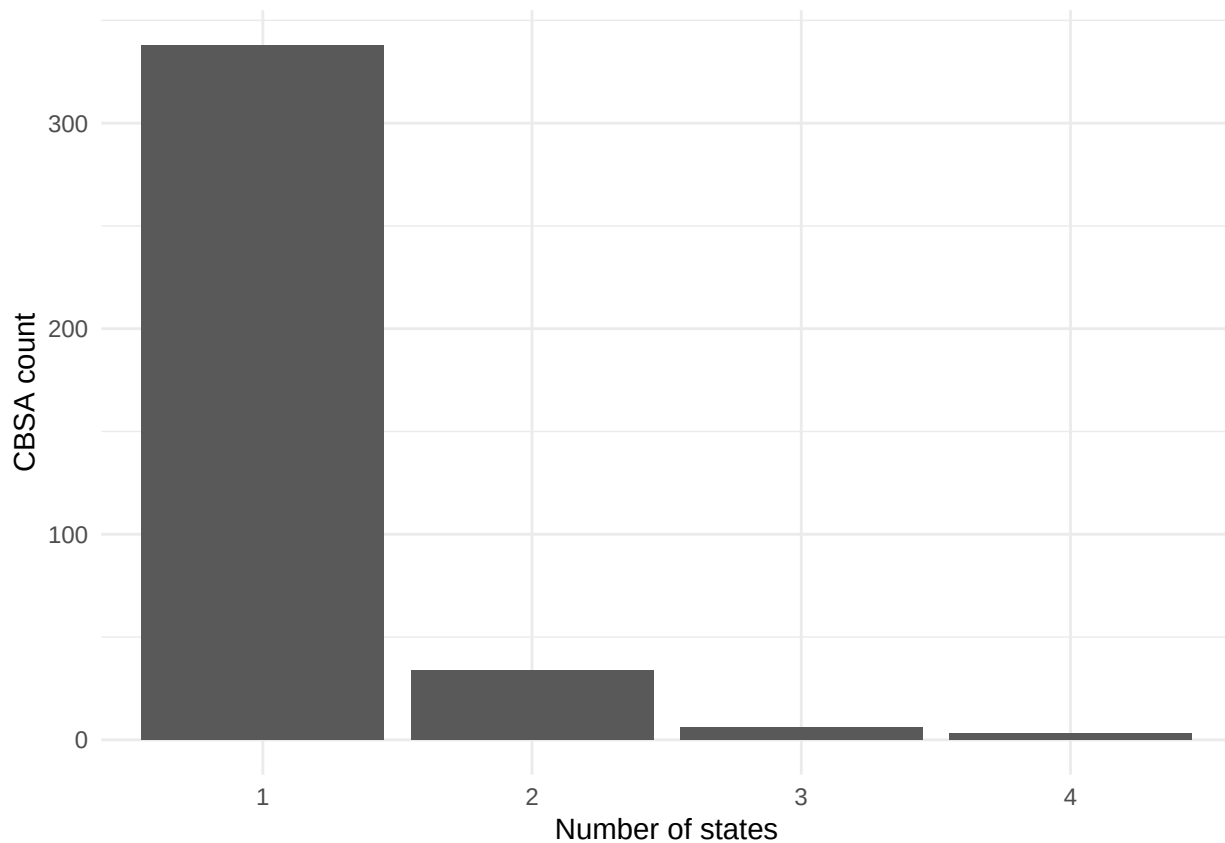
```
## [1] 1
```

Histogram of cbsa_data\$pm25



```
## # A tibble: 0 x 2
## # i 2 variables: geoid <fct>, n_cbsa <int>
```

```
## # A tibble: 43 x 2
##   cbsa_id n_states
##   <fct>     <int>
## 1 10900         2
## 2 12260         2
## 3 14140         2
## 4 16740         2
## 5 16860         2
## 6 16980         3
## 7 17140         3
## 8 17300         2
## 9 17980         2
## 10 19060        2
## # i 33 more rows
```



```
## NULL
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: cbsa_data
```

```
## Models:
```

```
## cc_model1: pm25 ~ pct_white * pct_pov + ruca_agg + density_log + (1 | state_abb/cbsa_id)
## cc_model: pm25 ~ pct_white * pct_pov + ruca_agg + density_log + (1 | cbsa_id) + (1 | state_abb)
##          npar    AIC    BIC logLik -2*log(L) Chisq Df Pr(>Chisq)
## cc_model1   10 138959 139049 -69470    138939
## cc_model    10 139191 139281 -69586    139171     0  0
```

```
## Some of the nested models seem to be identical and probably only vary in
## their random effects.
```

```
## # Comparison of Model Performance Indices
```

```
##
```

```
## Name      |          Model |    AIC (weights) |    AICc (weights)
```

```
## -----
```

```
## cc_model1 | lmerModLmerTest | 1.4e+05 (>.999) | 1.4e+05 (>.999)
```

```
## cc_model  | lmerModLmerTest | 1.4e+05 (<.001) | 1.4e+05 (<.001)
```

```
##
```

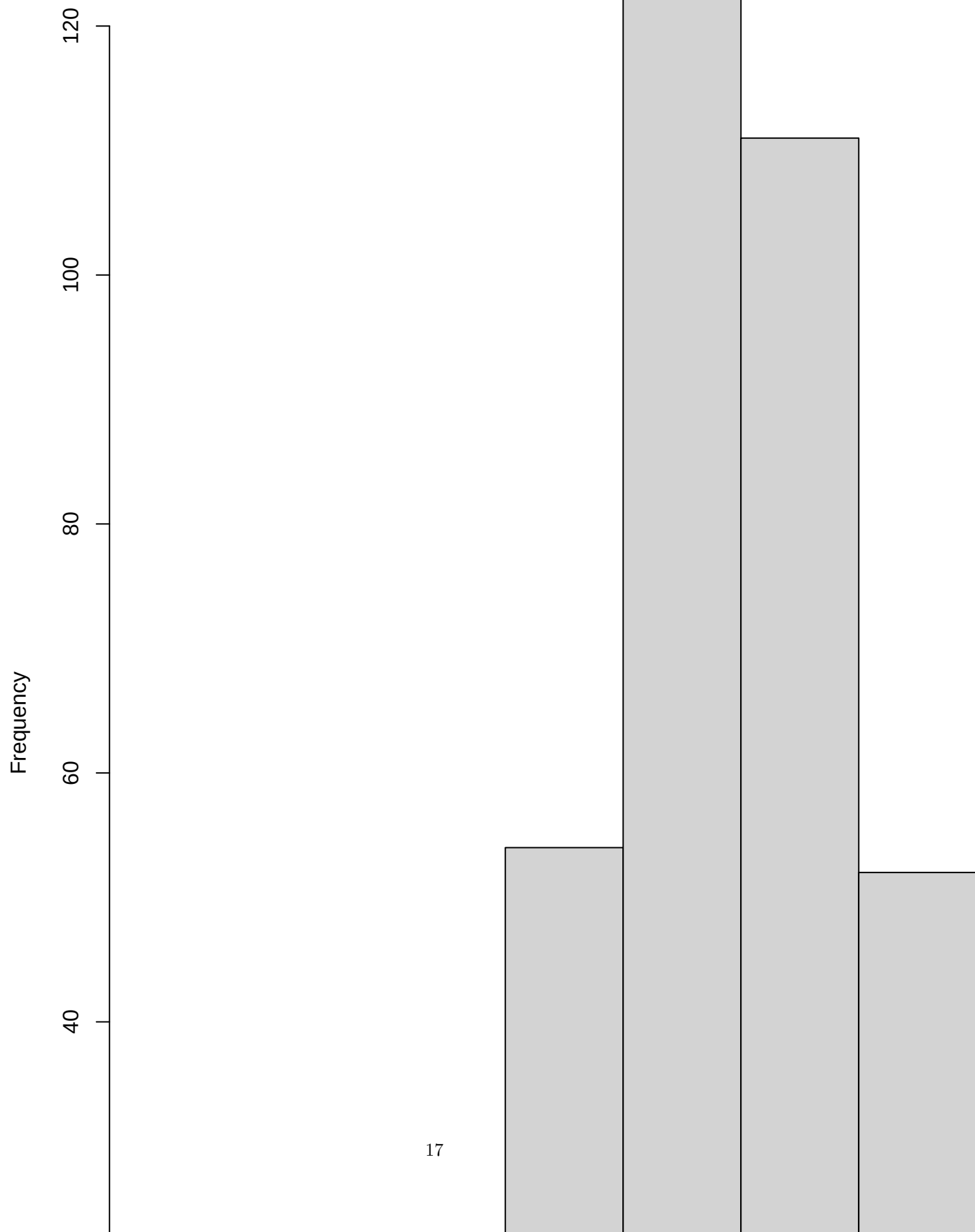
```
## Name      |    BIC (weights) | R2 (cond.) | R2 (marg.) |    ICC | RMSE | Sigma
```

```
## -----
```

```
## cc_model1 | 1.4e+05 (>.999) |    0.906 |    0.115 | 0.894 | 0.783 | 0.786
```

```
## cc_model  | 1.4e+05 (<.001) |    0.898 |    0.123 | 0.884 | 0.785 | 0.788
```

pdf
2



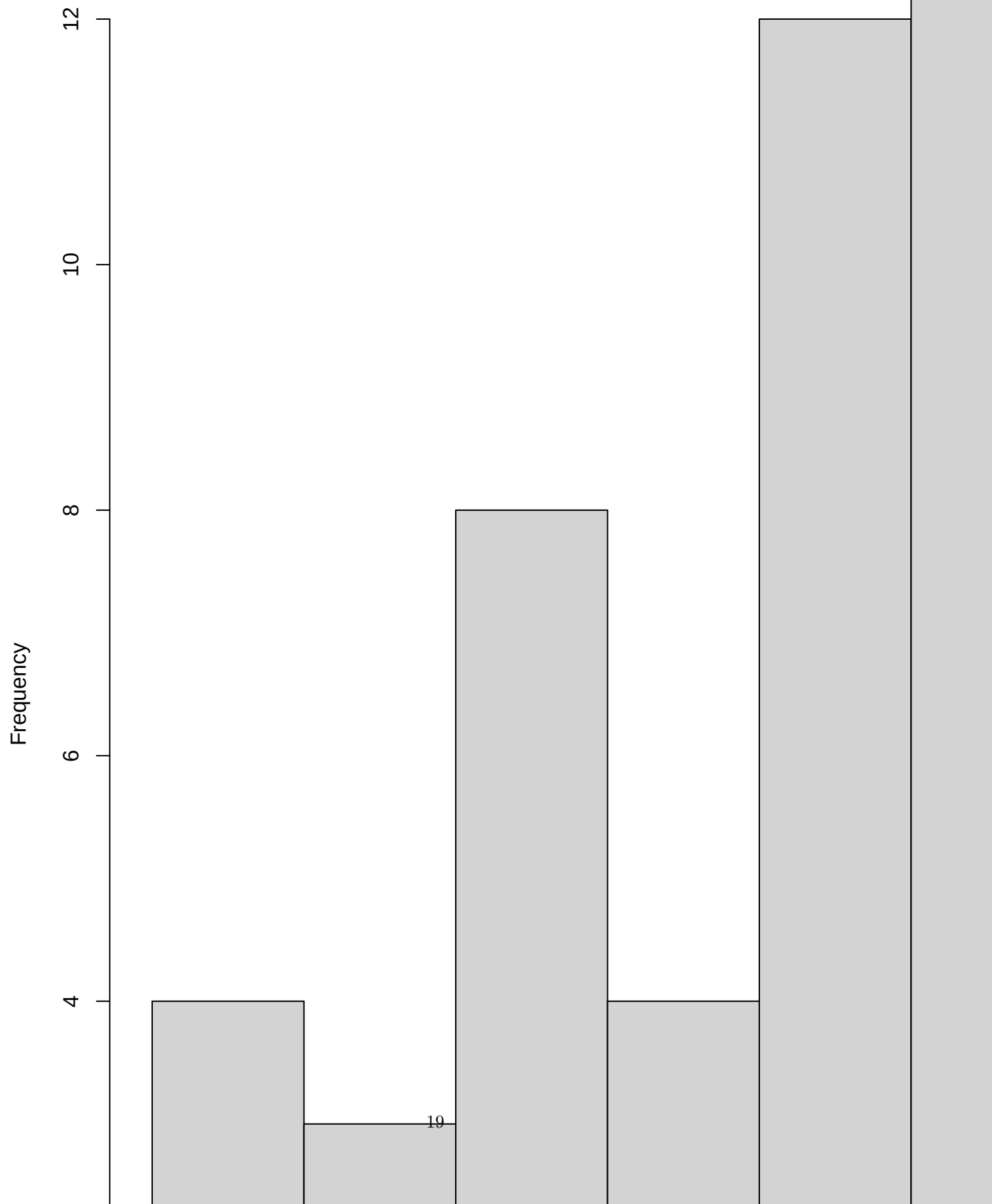


Table 3: Cross-Classified Mixed Effects Model for PM2.5

Term	Beta	95% CI	p-value
(Intercept)	6.4879	5.9392, 7.0365	<0.001
Percent White in pop	-0.0070	-0.0075, -0.0066	<0.001
Percent pop in pov	-0.0007	-0.0017, 0.0004	0.2
Rural-Urban Commuting Area			
rural	—	—	
suburban	-0.2369	-0.3004, -0.1733	<0.001
urban	0.3618	0.3102, 0.4134	<0.001
Log density (people/sq km)	0.4143	0.4091, 0.4194	<0.001
Percent White in pop * Percent pop in pov	0.0001	0.0001, 0.0001	<0.001

Abbreviation: CI = Confidence Interval

```
## # Indices of model performance
##
## AIC      | AICc | BIC | R2 (cond.) | R2 (marg.) | ICC | RMSE | Sigma
## -----
## 1.4e+05 | 1.4e+05 | 1.4e+05 | 0.898 | 0.123 | 0.884 | 0.785 | 0.788
```

```
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmprxES9W/file6c3e6793fb52.html screenshot
```

Approach 3: Multiway cluster adjusted SE (CRSE)

```
## NOTE: 1 observation removed because of NA values (LHS: 1).
```

```
## OLS estimation, Dep. Var.: pm25
## Observations: 58,029
## Standard-errors: Clustered (cbsa_id & state_abb)
##
##           Estimate Std. Error   t value   Pr(>|t|)
## (Intercept)    6.63421325    0.635446  10.440250 1.0158e-13 ***
## pct_white      -0.00567028    0.007538  -0.752221 4.5575e-01
## pct_pov         0.01616703    0.010325   1.565840 1.2424e-01
## ruca_aggsuburban 0.20523251    0.287207   0.714580 4.7848e-01
## ruca_aggurban    0.94486114    0.272596   3.466157 1.1550e-03 **
## density_log      0.34427771    0.071406   4.821389 1.5966e-05 ***
## pct_white:pct_pov -0.00000139    0.000128  -0.010864 9.9138e-01
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 2.14558   Adj. R2: 0.136367
```

```
## Warning: package 'data.table' was built under R version 4.5.2
```

```
##
## Attaching package: 'data.table'
```

```
## The following objects are masked from 'package:lubridate':
```

```
##
##   hour, isoweek, isoyear, mday, minute, month, quarter, second, wday,
##   week, yday, year

## The following objects are masked from 'package:dplyr':
##
##   between, first, last

## The following object is masked from 'package:purrr':
##
##   transpose

##               term cluster_cbsa cluster_state cluster_both
##               <char>         <num>         <num>         <num>
## 1:               (Intercept)      0.6577         0.6343         0.6354
## 2:      Percent White in pop      0.0077         0.0075         0.0075
## 3:      Percent pop in pov       0.0137         0.0102         0.0103
## 4:      ruca_aggsuburban        0.2278         0.2831         0.2872
## 5:      ruca_aggurban          0.2237         0.2686         0.2726
## 6: Log density (people/sq km)    0.0630         0.0696         0.0714
## 7:      White X poverty          0.0001         0.0001         0.0001

## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmprxES9W/file6c3e76c82c65.html screenshot

## [1] 253293.1

## [1] 253355.8

## [1] 139264.6

## [1] 139354.3

## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmprxES9W/file6c3e4fdad7cf.html screenshot
```

Approach 4: Generalized additive mixed models (GAMMs)

```
## refitting model(s) with ML (instead of REML)

## Data: NULL
## Models:
## linear_model$mer: NULL
## smooth_model$mer: NULL
##               npar      AIC      BIC logLik -2*log(L)  Chisq Df Pr(>Chisq)
## linear_model$mer    10 138959 139049 -69470    138939
## smooth_model$mer    11 138930 139028 -69454    138908 31.387  1 2.113e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## pdf
##   2
```

```
## 'log Lik.' -69484.55 (df=11)

## [1] 138991.1

## [1] 139089.8

## [1] 0.7859625

## Warning in tidy.lme(x, ...): ran_pars not yet implemented for multiple levels
## of nesting
## Warning in tidy.lme(x, ...): ran_pars not yet implemented for multiple levels
## of nesting

## x Unable to identify the list of variables.
##
## This is usually due to an error calling `stats::model.frame(x)` or `stats::model.matrix(x)`.
## It could be the case if that type of model does not implement these methods.
## Rarely, this error may occur if the model object was created within
## a functional programming framework (e.g. using `lapply()`, `purrr::map()`, etc.).

## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmprxES9W/file6c3e7290d706.html screenshot
## file:///var/folders/vs/qg7f_27x4b78c38y1jgv5m3w0000gq/T//RtmprxES9W/file6c3e31505ad.html screenshot
```

secondary outcome: BC

Characteristic	Beta	95% CI	p-value
(Intercept)	8.164	7.806, 8.523	<0.001
pct_white	-0.004	-0.004, -0.003	<0.001
pct_pov	-9.92e-04	-0.002, 8.07e-05	0.070
ruca_agg			
rural	—	—	
suburban	-0.272	-0.337, -0.207	<0.001
urban	0.017	-0.036, 0.069	0.5
log(dens_mult)	0.205	0.199, 0.210	<0.001
pct_white * pct_pov	6.17e-05	4.32e-05, 8.02e-05	<0.001

Abbreviation: CI = Confidence Interval